

VIVI

Voice-first vehicle search

A bounded conversational agent over grounded commercial-vehicle data

1,000 catalog rows

27 / 27 core

94.4% variants

Voice + text

The model understands the request. Application code owns filters, SQL, ranking, state, and factual validation.

Find the right commercial vehicle

Connected to localhost:8000

Idle

Hold to talk

Processing

Send

10:32 AM

You

I need a vehicle for intra-city goods delivery in Pune. Budget around 12 lakhs.

10:32 AM

Vivi

Understood. Looking for used commercial vehicles in Pune within ~₹12,00,000 for intra-city goods delivery.

10:33 AM

You

Diesel or CNG, payload above 1.5 tons.

10:33 AM

Vivi

Got it. Here are the best matching options.

Top matches

Make / Model	Year	Price	KM	Fuel	Payload / GVW	Body	City	Verification	Why this match?	
Tata Ace Gold CNG+	2022	₹8.25 L	48,000 km	CNG	1.6 T	Pick-up	Pune	Verified	Within budget, CNG, high mileage in city	>
Mahindra Bolero Maxitruck Plus	2021	₹9.75 L	62,000 km	Diesel	1.7 T	Pickup	Pune	Verified	Reliable, good payload capacity	>
Ashok Leyland Dost+	2020	₹7.90 L	55,000 km	Diesel	1.5 T	Pick-up	Pune	Verified	Affordable, easy maintenance	>

Play response

0:00 / 0:18

Current understanding

Budget	Up to ₹12,00,000	
Body type	Pick-up / Open Body	
Fuel	Diesel or CNG	
City	Pune	
Purpose	Intra-city goods delivery	

Updated: 10:33 AM

Latency

STT	182 ms
Understanding	241 ms
Search	312 ms
Response	276 ms
TTS	388 ms
Speech-end to first-audio	1,399 ms

Measured for last interaction

20-minute walkthrough + live demonstration

The four-turn live demo

1

Normal search

Chhota truck, 5 lakh ke andar, city delivery ke liye.

2

Correction

Nahi, diesel nahi, CNG.

3

Zero results

Add an impossible budget + body + city combination.

4

Grounded follow-up

Second one ka payload aur GVW kya hai?

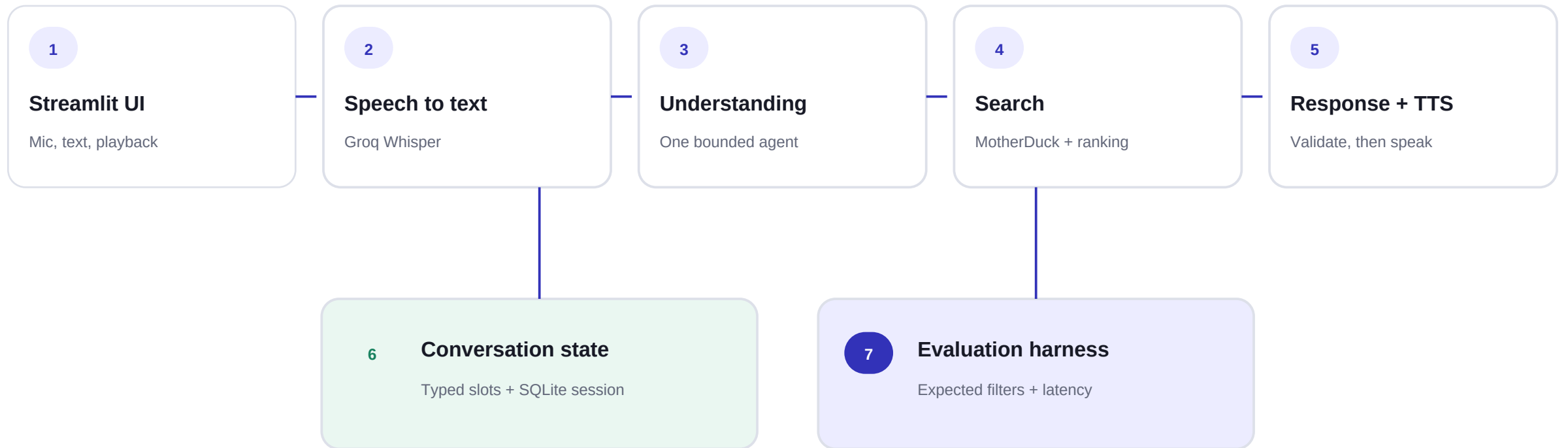
Visible slots

Grounded rows

Stage timings

Audio playback

Seven seams, one understandable process



The model interprets; code enforces

MODEL OWNS

- Intent selection
- Natural slot extraction
- Conversational wording

APPLICATION OWNS

- Typed tool schemas
- Parameterized SQL
- Hard filters + ranking
- State transitions

VALIDATOR OWNS

- Allowed vehicle IDs
- Allowed numeric facts
- Deterministic fallback

Invariant A vehicle fact can be conversationally rephrased, but every ID and numeric value must exist in the returned catalog records.

Conversation without a giant prompt

- Three typed operations: search, details, and catalog information.
- State carries active slots and prior result IDs across turns.
- Corrections patch only named fields; omitted fields remain unchanged.
- Pagination and all-result questions are explicit deterministic actions.
- General vehicle questions can be answered without fabricating catalog access.
- Out-of-scope or malicious requests are refused at the bounded tool/data boundary.

Why one agent?

The domain has three operations. A selector agent plus specialist agents would add latency, tokens, hand-offs, and more failure states without adding a real capability.

Explainable

Replaceable seams

Evaluation: broad cases, executable expectations

27 / 27

Core live cases

94.4%

Full variant run

2,144

Mean tokens / turn

INR 0.046

Mean LLM list cost

1,794 ms

Core mean text turn

What the executable sets cover

- Every size class and body variant
- Fuel, city, budget, payload, GVW
- More options, all weights, all details
- Expected action, slots, records, and meaning

One miss, diagnosed instead of hidden

The full variant run scored 17/18. The answer said 'listed in Delhi'; the concept checker recognized only city, located in, or location. No fact, action, filter, or result ID was wrong. The corrected two-turn check passed 2/2.

Every run saves JSON plus a timestamped Markdown report under data/evaluation/.

Trade-offs I can defend

CHOSEN	REJECTED	WHY HERE
Typed tools + deterministic SQL	Model-generated SQL	Strict constraints and a smaller data boundary
One bounded agent	Agent hierarchy	Less latency, tokens, state transfer, and code
Validated natural response	Prompt-only or rigid template	Natural speech with code-level factual checks
Cascaded STT / agent / TTS	Speech-to-speech	Inspectable transcript, slots, and timings
Scoped DB connection	Process-global connection	Clear ownership across concurrent sessions

At 100k conversations/month, capacity breaks first

- 1 Provider capacity** Paid capacity, circuit breakers, p95 and cost telemetry
- 2 Voice latency** Stream STT/TTS; keep simple searches on one model call
- 3 Session concurrency** Move SQLite state to Redis with TTL and size limits
- 4 Catalog access** Bounded pool/read service, indexes, option caching
- 5 Model drift** Pinned CI eval plus separate provider canary set

Measured core mix: INR 0.046 LLM list cost / text turn; about INR 4,605 at 100k turns.

Voice, database, hosting, retries, and production discounts remain separate.

Demo ready. Questions welcome.

Live checklist

- Normal voice search
- One-slot correction
- Zero-result relaxation
- Prior-result detail follow-up
- Show slots, rank data, timing, and tokens

Useful Q&A entry points

- Why not model-generated SQL?
- What happens after an STT mishear?
- Show ranking evidence, not prose
- Where would you win the first 200 ms?
- What fails under concurrent load?

Sources: docs/SOURCES.md | Evaluation evidence: docs/EVALUATION.md

The boundary is the product: conversational outside, deterministic around catalog facts.